Lecture 09: Multiples & Technical Analysis

Questions

Choose one answer. Keep full precision until the final result. Worked solutions follow the complete question set.

1. Trailing versus forward P/E

A share trades at \$120. EPS over the latest four reported quarters is \$4, and expected EPS over the next twelve months is \$5. What are its trailing and forward P/E ratios, respectively?

A. 30x and 24x
B. 24x and 30x
C. 30x and 30x
D. 24x and 24x

2. From $EV/EBITDA$ to equity per share

All amounts are in \$ millions except per-share values. EBIT is 80 and $D\&A$ is 20. An appropriate $EV/EBITDA$ multiple is 9x. Financial debt is 260, nonoperating cash is 60, and there are 35 million shares. Assume no other claims or adjustments. What is the implied price per share?

A. \$20.00
B. \$25.71
C. \$17.71
D. \$14.86

3. EV/Sales still needs the equity bridge

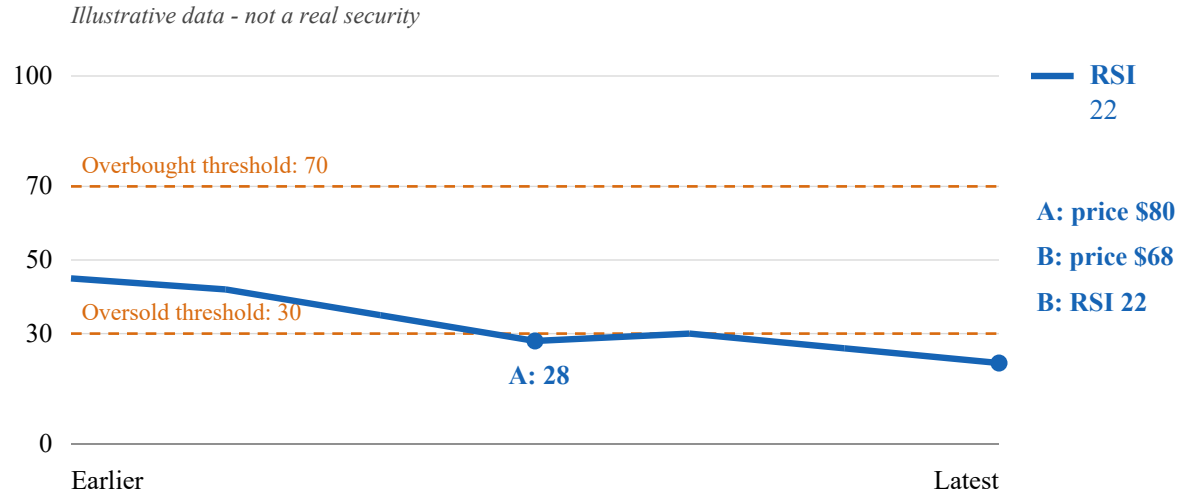
A firm has annual sales of \$800 million. Apply an EV/Sales multiple of 2.5x. Financial debt is \$500 million, nonoperating cash is \$100 million, and there are 40 million shares. Assume no other claims or adjustments. What is the implied price per share?

A. \$40.00
B. \$65.00
C. \$50.00
D. \$37.50

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4. Oversold is not a guaranteed floor

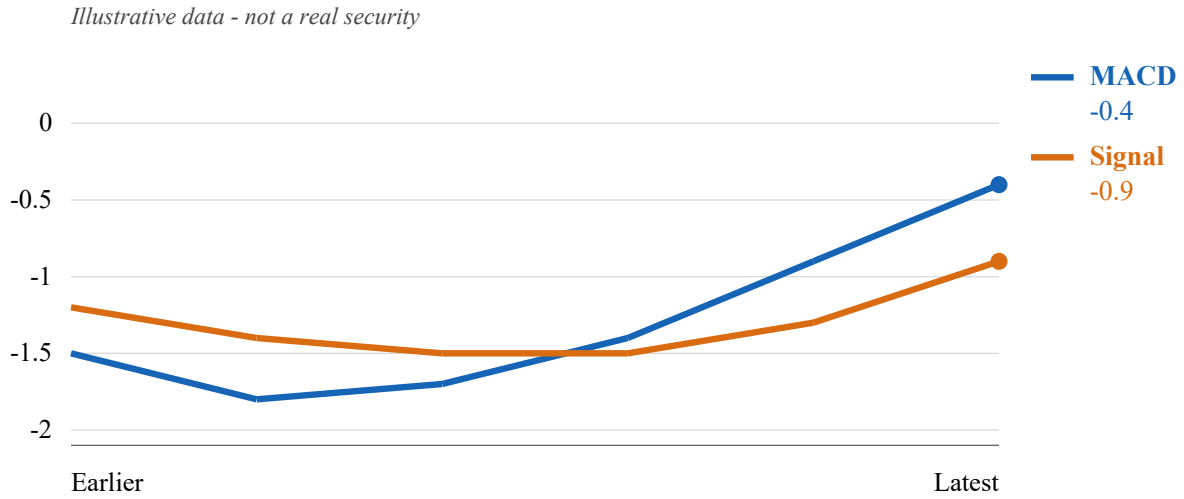
At date A, a stock trades at \$80 with $RSI = 28$. At date B, it trades at \$68 with $RSI = 22$. Which statement correctly combines the price change and the RSI interpretation?



A. The stock fell 15%; RSI below 30 indicated oversold momentum conditions, not a guaranteed rebound.
B. The stock fell 12%; the RSI decline measured the percentage price decline.
C. The stock rose 15%; a falling RSI is always bullish.
D. The stock fell 17.65%; RSI below 30 guaranteed a rebound.

5. MACD and its signal line

At the latest date in the illustrative chart, MACD is -0.40 and its signal line is -0.90. Without calculating the moving averages, which reading is most accurate?



A. MACD is below its signal line, so the histogram must be negative.
B. MACD is above its signal line but below zero: momentum has improved relative to the signal, while the fast EMA remains below the slow EMA.
C. MACD above its signal line guarantees a positive return next period.
D. MACD below zero means the stock has negative intrinsic value.

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Worked solutions

1. Trailing versus forward P/E

Answer A. 30x and 24x

Step 1. Match each denominator to its time period.

Read the time labels before calculating anything.
Trailing earnings have already been reported.
Forward earnings are a forecast.
We are comparing the same current share price with two different earnings measures.

Step 2. Use the same current price in both ratios.

Divide the current price by each EPS figure.
Since forecast earnings are higher, the forward multiple is lower.
Nothing about this calculation requires the price to change.

$$(P/E)_{\text{trailing}} = \frac{120}{4} = 30, \qquad (P/E)_{\text{forward}} = \frac{120}{5} = 24$$

Step 3. Explain why the ratios differ.

I would not conclude that the stock is cheap just because the forward multiple is lower.
The forecast could be too optimistic.
When using peers, match forward multiples with forward earnings and trailing multiples with trailing earnings.

2. From $EV/EBITDA$ to equity per share

Answer A. $20.00

Step 1. Construct the EBITDA proxy used in class.

First build the EBITDA measure used in class.
EBIT has already deducted depreciation and amortization, so add that expense back.
Keep all amounts in millions throughout the calculation.

$$EBITDA = EBIT + D\&A = 80 + 20 = 100$$

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Step 2. Value the operating business, then bridge to equity.

The multiple gives us enterprise value, which is the value of the operating business.
We then move to the equity claim: subtract financial debt and add nonoperating cash.
The question rules out other claims, so no further adjustments are needed.

$$EV = 9 \times 100 = 900$$
$$V_E = EV - D + \text{Cash} = 900 - 260 + 60 = 700$$

Step 3. Convert the total claim into a share price.

Only after that bridge do we divide by the number of shares.
Millions of dollars divided by millions of shares gives dollars per share.
Dividing enterprise value directly by shares would skip the lenders' claim.

$$P_0 = \frac{V_E}{N} = \frac{700}{35} = \boxed{\$20}$$

3. EV/Sales still needs the equity bridge

Answer A. $40.00

Step 1. Use the multiple to obtain enterprise value.

The sales multiple is an enterprise-value multiple.
Multiply annual sales by the chosen multiple to value the operating business.
At this stage, we have not yet isolated what belongs to shareholders.

$$EV = (EV/\text{Sales}) \times \text{Sales} = 2.5 \times 800 = 2{,}000$$

Step 2. Remove debt and add nonoperating cash.

Now subtract financial debt and add nonoperating cash.
Notice that cash has the opposite sign from debt.
With no other claims or adjustments in the question, this gives total equity value.

$$V_E = EV - D + \text{Cash} = 2{,}000 - 500 + 100 = 1{,}600$$

Step 3. Express the estimate per share.

Divide total equity value by shares to get the price per share.
The calculation is simple, but selecting a defensible sales multiple is the harder economic judgment: profitability and growth still matter.

$$P_0 = \frac{1{,}600}{40} = \boxed{\$40}$$

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4. Oversold is not a guaranteed floor

Answer A. The stock fell 15%; RSI below 30 indicated oversold momentum conditions, not a guaranteed rebound.

Step 1. Calculate the actual price change.

First calculate what actually happened to the share price.
Start with the initial price of $80 and compare it with the final price of $68.
Divide by the initial price, not the final price.
This gives the percentage return over the interval.

$$R_{A\to B} = \left(\frac{P_B}{P_A} - 1\right) \times 100$$

$$= \left(\frac{68}{80} - 1\right) \times 100 = \boxed{-15\%}$$

Step 2. Read the indicator in its own units.

Now read RSI separately.
At date A, RSI is 28, already below the usual oversold threshold of 30.
At date B it falls to 22.
These are indicator readings on a scale from zero to 100; they are not percentage returns.
The move from 28 to 22 is a six-point change in RSI, not the stock's return.

Step 3. Draw the right lesson.

The key point is that the stock kept falling after it was already oversold.
I would use RSI as information about recent momentum, not as a promise that a rebound is about to happen.
A low RSI also does not establish that the stock is below intrinsic value.
We still need a valuation model to make that judgment.

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5. MACD and its signal line

Answer B. MACD is above its signal line but below zero: momentum has improved relative to the signal, while the fast EMA remains below the slow EMA.

Step 1. Compare the two plotted lines.

With negative numbers, compare their positions on the number line.
Minus 0.40 is above minus 0.90.
MACD is therefore above its signal line, and their difference is positive.

$$-0.40 > -0.90$$

$$\text{Histogram} = MACD - \text{Signal} = -0.40 - (-0.90) = +0.50$$

Step 2. Then compare MACD with zero.

The zero line is a separate comparison.
MACD itself is still negative, so the faster exponential moving average remains below the slower one.
A positive histogram does not mean MACD must also be positive.

Step 3. Keep both pieces of information.

My interpretation would be improving momentum relative to the signal, while the fast average has not yet moved above the slow average.
That is useful context, but it is neither a valuation estimate nor a guarantee about the next return.